DC Charging System (BEVC-DC001)-I

Introduction

As the world transitions towards sustainable transportation, the development of robust and efficient electric vehicle (EV) charging infrastructure becomes paramount. In India, this critical need is addressed by standards like the Bharat Electric Vehicle Charging (BEVC) DC001. This lecture delves into the specifics of the BEVC-DC001 standard, a foundational DC charging system designed to support the growing fleet of low-voltage powertrain EVs, including two-wheelers, three-wheelers, and early-stage four-wheelers. Understanding this standard is crucial for anyone involved in the design, deployment, or operation of EV charging solutions in the Indian context, as it outlines the technical specifications, communication protocols, and operational requirements that ensure safe and effective DC fast charging.

This session builds upon previous discussions of CCS2 DC charging systems, highlighting the unique aspects and communication methodologies employed by BEVC-DC001. A significant focus is placed on the Controller Area Network (CAN) communication, which forms the backbone of data exchange between the EV and the charging equipment (EVSE) in this standard. By exploring the intricacies of CAN communication, its benefits, and its implementation within BEVC-DC001, we gain a comprehensive understanding of how these systems operate and contribute to the broader EV ecosystem.

Core Concepts

Term	Definition	Significance
`BEVC-DC001`	The Bharat Electric Vehicle Charging standard DC001, introduced by the Department of Heavy Industries, Government of India in 2017.	It is an initial and still prevalent DC fast charging standard in India, primarily supporting low-voltage powertrain EVs.
`CAN Communication`	Controlled Area Network, a message-based, two-wire serial bus communication protocol used for data exchange between electronic control units.	It is the dedicated digital communication method between the EV and EVSE in BEVC-DC001, enabling efficient and robust data sharing.

Term	Definition	Significance
`GBT 27930`	A Chinese national standard for EV conductive charging communication protocols, specifically GB/T 27930-2015.	BEVC-DC001 derives its communication rules, including application layer parameters and Protocol Data Unit (PDU) formats, from this standard.
`SAEJ1939`	A set of standards defined by the Society of Automotive Engineers (SAE) for vehicle networks, particularly for heavy-duty vehicles.	The physical and data link layers of BEVC-DC001's CAN communication are derived from SAEJ1939, ensuring robust low-level communication.
`Proximity Pilot (PP)`	A pin in charging connectors (like CCS2) that detects the physical connection of the charger plug to the vehicle and immobilizes the vehicle.	While CCS2 uses PP, BEVC-DC001 uses dedicated charging confirmation pins for hardware-based physical connection detection and locking.
`Control Pilot (CP)`	A pin in charging connectors that communicates charging parameters (voltage, frequency, duty ratio) between EV and EVSE.	In CCS2, CP handles AC charging parameters and DC charging states; BEVC-DC001 uses CAN communication for these functions.
`Mode 4 Charging`	A charging mode where a dedicated DC Electric Vehicle Supply Equipment (EVSE) performs DC charging of the vehicle.	BEVC-DC001 is a Mode 4 charging system, indicating that the power conversion unit is external to the vehicle, within the EVSE.
`PDU (Protocol Data Unit)`	A specific format for CAN data frames, derived from GB/T 27930, used to transmit messages between the EV and EVSE.	This format allows controllers to identify and process messages, facilitating high-level communication of charging parameters.

Term	Definition	Significance
`EVSE`	Electric Vehicle Supply Equipment, the charging station or device that supplies electrical energy to charge an EV.	BEVC-DC001 defines specific types of EVSEs (Type 1 and Type 2) with varying power outputs and connector configurations.

Detailed Analysis

1. Evolution and Context of DC Charging Systems

The landscape of DC charging systems has evolved significantly, with various standards addressing different regional and vehicle-specific needs. Previously, the CCS2 DC charging system was discussed, which involves complex communication sequences, including proximity pilot (PP) for vehicle immobilization, control pilot (CP) for AC charging parameters and DC charging states, and Power Line Communication (PLC) for high-level data exchange. In CCS2, PLC occurs over the CP and PE lines, enabling the exchange of battery and charging state information between the vehicle and the external power conversion unit.

In contrast, the BEVC-DC001 standard, introduced by the Department of Heavy Industries, Government of India in 2017, represents a distinct approach, particularly for low-voltage powertrains common in two-wheelers, three-wheelers, and early four-wheelers in India. While CCS2 relies on PLC for high-level communication, BEVC-DC001 utilizes a dedicated Controller Area Network (CAN) communication channel. This difference in communication methodology is a key distinguishing factor, alongside the specific power output capabilities and connector types tailored for the Indian market. Both standards, however, share the fundamental principle of Mode 4 charging, where the power conversion unit resides within the EVSE, necessitating continuous information exchange between the EV and EVSE.

2. BEVC-DC001 Standard Specifications

The BEVC-DC001 standard is designed with specific input and output characteristics to cater to its target vehicle segments.
* **Input**: It accepts a three-phase AC input of 415 volts (line-to-line), utilizing a five-wire connection (three phase wires, one neutral, and one earth wire).
* **Output**: The standard supports a maximum output of 10 kW for 48V systems, corresponding to approximately 200A. For 72V systems, it can support up to 15 kW, also maintaining a maximum current of around 200A.

Connector and Pinout:
The standard specifies the use of a GB/T 20234.3 connector with a standard 5-meter cable. The pin configuration of this connector is crucial for its operation:
* **DC+ and DC-**: These are the primary power transfer pins, designed to be thicker due to the high current flow.
* **Charging Confirmation Pins (CC1, CC2)**: These pins provide a hardware-based confirmation of the charger plug's physical connection and ensure its locking into the vehicle inlet.
* **CAN Communication Pins (S+, S-)**: S+ corresponds to the CAN high line, and S- to the CAN low line. These pins facilitate the actual CAN communication for data exchange between the EV and EVSE.
* **Auxiliary Pins (A+, A-)**: These pins supply power to auxiliary systems within the charger, such as controllers and sensors.
* **Protective Earth (PE)**: For safety grounding.

EVSE Types and Power Ratings:
The standard defines two types of EVSEs:
* **Type 1 EVSE**:
* Output 1: Limited to 10 kW for 48V or 72V systems.
* Output 2: Limited to 3.3 kW for 48V systems, primarily for two-wheelers.
* **Type 2 EVSE**:
* Output 1: Provides 10 kW for 48V systems and 15 kW for 72V systems (both up to 200A).
* Output 2: Same as Type 1 Output 2, 3.3 kW for 48V two-wheelers.
All Output 1 connectors for both types are derived from the GB/T 20234.3 standard, ensuring consistent dimensions and proximity detection.

Functional Requirements and Protection:
The standard mandates specific performance criteria for EVSEs:
* **Measurement Accuracy**: Voltage measurements must be within ±0.5%, and current measurements within ±1 ampere for current levels ≤50 amperes.
* **Overvoltage Protection**: If the output voltage exceeds the maximum battery voltage limit for 1 second, the EVSE must reduce the DC output current to less than 5 amperes within 2 seconds to prevent battery damage.
* **Communication**: Dedicated CAN communication between EV and EVSE, with properties incorporated from IEC 61851-24 and GB/T 27930 (specifically AIS 138 Part 2 Annex G). Communication between the EVSE and a central management system uses the Open Charge Point Protocol (OCPP) for metering, logging, and maintenance.
* **Regulation Method**: A combination of constant voltage (CV) and constant current (CC) control methods.
* **Power Conversion Unit Characteristics**: Overall efficiency must be greater than 92% at nominal power output, and the power factor must be greater than 0.9

during full load.
* **Current Limit**: Maximum current limit is 200 amperes.
* **Regulation Tolerances**:
* Constant Current (CC) mode: ±2.5 amperes for current levels <50 amperes, and ±5% for current levels >50 amperes.
* Constant Voltage (CV) mode: Maximum 2% voltage tolerance for the maximum rated EVSE voltage.

3. Communication in BEVC-DC001

Digital communication in BEVC-DC001 is exclusively handled by the Controlled Area Network (CAN). This dedicated communication channel ensures reliable data exchange between the EV and EVSE.
* **CAN Bus Lines**: It uses dedicated CAN high (CAN_H) and CAN low (CAN_L) lines, implemented with shielded twisted pair wires and a termination resistance of 120 ohms.
* **Proximity Detection**: Unlike the proximity pilot in CCS2, BEVC-DC001 uses charging confirmation pins (CC1, CC2) for hardware-based detection of physical connection and locking of the charger plug. Digital communication commences only after this physical confirmation and locking.
* **Layered Communication Model**: The communication protocol is derived from GB/T 27930-2015, which defines rules for implementing various layers:
* **Physical Layer**: Derived from SAEJ1939-11/ISO 11898-1, it defines termination conditions, voltage levels, differential signaling, and communication speed.
* **Data Link Layer**: Derived from SAEJ1939-21, it defines addressing for communication nodes (BMS of EV and charger controller) and CAN data frame formats.
* **Application Layer**: Taken from GB/T 27930-2015, it defines the representation of parameters, their parameter group numbers, and specific messages, including data units and resolutions. Some aspects are customized for the Indian region.
* **Data Transmission**: CAN data frames are transmitted in a Protocol Data Unit (PDU) format, enabling controllers to identify and process messages.
* **Speed and Protocol**: The communication speed is 250 kbps, utilizing the CAN 2.0b protocol, which features a 29-bit identifier.
* **Dedicated Participants and Addresses**: Communication involves only two participants: the charger controller and the vehicle's Battery Management System (BMS). They have fixed addresses:
* Charger: 86 (decimal) or 56 (hexadecimal)
* BMS: 244 (decimal) or F4 (hexadecimal)
This communication channel is entirely separate from other in-vehicle CAN systems (e.g., powertrain CAN), ensuring dedicated and secure data flow for charging.

Introduction

CAN (Controlled Area Network) is a robust, message-based communication network designed for efficient data exchange between electronic control units (ECUs) or devices. Introduced by Bosch at the 1991 SAE conference, it quickly became a global standard for in-vehicle communication.
* **Message-Based Protocol**: In CAN, the priority of a message is embedded within the message itself (via its ID). This allows for flexible network expansion or modification by simply changing software, as nodes can be added or removed without extensive hardware reconfigurations.
* **Two-Wire Serial Bus**: CAN operates on a two-wire system (CAN high and CAN low) using differential signaling. Messages are sent serially, bit by bit, over this single pair of wires.
* **Purpose**: The primary motivation for CAN's development was to reduce the extensive wiring harnesses within vehicles. Instead of multiple dedicated wires for each data signal between controllers, CAN allows numerous controllers to share information over a single pair of wires.
* **Applications**: Widely adopted in the automotive industry, CAN is also prevalent in avionics, medical systems, and industrial automation (e.g., escalators).

Wiring Reduction Example:
Consider a vehicle with an Engine Control Unit (ECU), a propulsion system, an Anti-lock Braking System (ABS), and an infotainment system, all requiring speed data from a single speed sensor.
* **Without CAN**: Each controller would need a dedicated wire (or pair of wires) from the speed sensor. For four controllers, this could mean eight copper wires running throughout the vehicle, increasing weight, cost, and complexity.
* **With CAN**: The speed sensor connects to the CAN bus via a CAN transceiver. The speed information is then broadcast onto the CAN high and CAN low lines. All four controllers, also connected to the same CAN bus, receive this information simultaneously. Each controller then decides the relevance of the message based on its internal CAN database. This reduces the wiring from eight wires to just two, significantly simplifying the vehicle's electrical architecture.

5. Benefits of CAN Communication

The widespread adoption of CAN communication stems from its numerous advantages:
* **Low Cost**: By reducing the amount of copper wiring required for inter-controller communication, CAN significantly lowers manufacturing costs and vehicle weight.
* **Centralized Communication**: It enables central diagnosis and configuration of all connected controller units, streamlining maintenance and system management.

* **Robustness**: CAN signals are highly resistant to electromagnetic interference (EMI) from nearby circuits. This is achieved through differential signaling and the use of twisted pair cables, making it suitable for electromagnetically noisy environments.
* **Efficiency**: Messages are prioritized by their IDs. The message with the highest priority gains immediate access to the CAN bus, ensuring critical data is transmitted without delay.
* **Error Capability**: CAN data frames include a Cyclic Redundancy Check (CRC) code, which performs error checking on each frame. If an error is detected, an error frame is transmitted, and all nodes discard the faulty message, ensuring data integrity.
* **Flexibility**: Each CAN-connected controller receives all messages transmitted on the bus. It then determines the relevance of a message by comparing its ID with its internal CAN database. This design allows for easy modification or the addition of new nodes in the future with minimal software changes.
* **Speed**: CAN supports data rates of 1 Mbps over cables up to 40 meters and 125 kbps over distances up to 100 meters.
* **Multi-Master Configuration**: Any node connected to the CAN bus can initiate communication if the bus is free. The "master" role dynamically shifts to the node transmitting the highest priority message, allowing for distributed control and efficient bus arbitration.

Key Takeaways

- [ ] BEVC-DC001 is an Indian DC charging standard for low-voltage EVs, introduced in 2017.
- [ ] It supports up to 10 kW for 48V systems and 15 kW for 72V systems, with a maximum current of 200A.
- [ ] The standard uses a GB/T 20234.3 connector and relies on dedicated CAN communication for data exchange.
- [ ] Communication protocols are derived from IEC 61851-24B and GB/T 27930, with physical/data link layers from SAEJ1939.
- [ ] BEVC-DC001 employs a three-layer communication model (Physical, Data Link, Application) over a dedicated CAN bus.
- [ ] The CAN communication operates at 250 kbps using CAN 2.0b protocol with 29-bit identifiers.
- [ ] Fixed CAN addresses are assigned to the charger (86 decimal) and the vehicle's BMS (244 decimal).
- [ ] CAN communication significantly reduces wiring complexity, cost, and weight in vehicles.

- [ ] Key benefits of CAN include robustness, efficiency (priority-based access), error checking (CRC), and flexibility.
- [ ] EVSEs under BEVC-DC001 must meet specific functional requirements for measurement accuracy, overvoltage protection, and efficiency.

Conclusion

The BEVC-DC001 standard plays a pivotal role in establishing a reliable and efficient DC charging infrastructure for electric vehicles in India. By defining clear specifications for power delivery, connector types, and, crucially, communication protocols, it ensures interoperability and safety across the charging ecosystem. The adoption of CAN communication as the backbone for data exchange between the EV and EVSE highlights a strategic choice for robustness, efficiency, and flexibility, drawing upon established automotive communication standards like GB/T 27930 and SAEJ1939.

Understanding the intricacies of BEVC-DC001, from its power output capabilities and EVSE types to its layered communication architecture and the underlying principles of CAN, is fundamental for engineers and stakeholders in the EV sector. This knowledge not only facilitates the development and deployment of compliant charging solutions but also provides a foundation for future innovations in smart charging and grid integration. As the EV market continues to expand, standards like BEVC-DC001 will remain critical enablers, ensuring that the charging experience is seamless, safe, and scalable.